

ASHWANTHRAM K L

+918681849991 k.l.ashwanthram@gmail.com ashwanthram AshwanthramKL

TECHNICAL SKILLS

Programming Languages: Python, Javascript
Frameworks & Libraries: FastAPI, MCP, Google ADK, PydanticAI, Langchain, LlamaIndex, Streamlit, Sentence Transformers
LLM Development: Prompt Engineering (Reflexion, Chain-of-thought, In-Context Learning, ReAct, Meta Prompting, Multi-Agent Debate), Fine-tuning (Supervised, DPO, Distillation), Evaluation (DeepEval, Latest Research)
MLOps & LLMOps: Monitoring & Observability (Pydantic Logfire, LangTrace)
AI/Machine Learning: Retrieval Augmented Generation (RAG), Semantic Search, Information Retrieval, STT and TTS(Elevenlabs, smallest.ai), Multimodal AI, DocumentAI (Llaparse, MistralOCR), NumPy, Pandas, Matplotlib
VectorDBs & Development Tools: FAISS, ChromaDB, Qdrant, MongoDB, Docker, REST APIs, Git, Github, Cursor, Windsurf

EXPERIENCE

ProdGain Pvt Ltd

06/2024 – Present

AI Engineer

Hyderabad

- **Architected a cutting-edge DeepResearch system** using **Claude 3.7 Sonnet**(primary) and **GPT-4o**(fallback), implementing a multi-step iterative methodology that executes targeted queries through **Exa** (primary) and **Brave** (fallback) search engines, enabling sophisticated market research with **domain filtering capabilities** that **outperformed ChatGPT's and Grok's Deep Research** in expert evaluations.
- **Implemented MQAG eval framework** that compares **probability distributions** over answer choices rather than text spans, conducting extensive evaluations that showed **Gemini Flash 2.0** achieved superior performance (**0.83 coverage, 0.98 alignment**) compared to GPT-4o-mini (**0.75 coverage, 0.97 alignment**) for summary quality assessment.
- **Developed a sophisticated search pipeline** with advanced relevance determination that analyzes search results against **topic-specific criteria**, ensuring content adheres to **domain inclusion/exclusion rules** (e.g., "don't consider blogs", "only consider government sources")—a core requirement specified by market research professionals.
- **Engineered comprehensive cost optimization procedures** through rigorous model benchmarking, implementation of Claude's **text editor tool** for efficient string replacements of drafts on follow-ups, and **intelligent content caching**, reducing per-DeepResearch costs from **\$2.50 to \$1.00-1.50** while maintaining superior research quality.

FarmwiseAI Pvt Ltd

08/2023 – 05/2024

Data Scientist Intern (GenAI)

Chennai

- **Constructed a topic tagging based recommendation system** using LLMs for a text-based social media platform, enhancing user engagement by accurately tagging posts and user profiles.
- **Implemented content routing** based on topic tags to increase user participation in discussions by **25%**.
- **Formulated and standardized Canonical Lists of Topics (CLOTs)** for 10+ spaces by integrating Langchain's SerpAPI Agent and LLMs for relevance scoring the topics, **improving tagging accuracy by more than 30%**.

PROJECTS

Smart Recruit: AI-Powered Recruitment Copilot | [Demo](#)

- **Designed a recruitment copilot** using Gemini models and **GTE-large embeddings** (512 token window), implementing a **two-stage candidate selection process** with semantic search in ChromaDB followed by **LLM-based re-ranking** that evaluates technical skills and experience relevance, **reducing initial screening time from 120+ mins to under 3 mins**.
- **Developed an AI interviewer** with structured prompt engineering for hallucination mitigation, using **WebSocket streaming** (500ms latency) with **Elevenlabs conversational agent** and **Twilio** integration, enabling 5-10 minute telephonic interviews with **personalized questions and dynamic follow-ups** that adapt to candidate responses while maintaining adherence to the predefined interview structure.
- **Implemented a customizable technical assessment system** that generate candidate-specific evaluation questions based on their claimed skills and job requirements, paired with an **evaluation agent** trained provide **quantitative scoring across multiple dimensions** including code correctness, problem-solving approach, and depth of knowledge.

All-in-One Chunker: Semantic PDF Processing System | [Demo](#)

- **Engineered a custom semantic chunker** using **Gemini models** that intelligently segments documents based on meaning rather than arbitrary limits, preserving context across page boundaries and improving RAG performance with a **15% increase in accuracy and 8% boost in recall** compared to traditional semantic chunking methods.
- **Designed comprehensive metadata extraction** that identifies entities, topics, keywords, potential search queries, etc. for each chunk, enhancing retrieval capabilities by additional **metadata filtering** beyond standard similarity search.
- **Implemented a cost-effective processing pipeline** with context caching that achieved comparable performance to LlamaParse's Premium tier while **reducing costs by approximately 49%**.

EDUCATION

Coimbatore Institute of Technology, Coimbatore

07/2020 – 05/2024

Bachelor of Engineering (B.E) Computer Science and Engineering